

Varun Uday Chebbi

FAE – Industrial AI and Digital Twin
(OT–IT Systems)

varunudaychebbi@gmail.com

<https://www.linkedin.com/in/varun-chebbi-065686192/>

+49-17648981740

Große parower straÙe 133, Stralsund, 18435, DE

Consultant – Industrial Artificial Intelligence & New Technologies with 5+ years of experience embedding AI solutions directly into operational and production environments. Proven expertise in Predictive Maintenance, Industrial IoT Analytics, Digital Twins, and Edge AI, bridging OT–IT architectures and translating complex industrial challenges into scalable, reliable AI applications. Strong consulting mindset with hands–on experience in Python–based machine learning, cloud AI platforms (Azure/AWS), industrial time–series data, and stakeholder communication across all seniority levels.

Work Experience

Research Assistant

Feb 2025 – Present

Hochschule Stralsund | Germany

- Led consulting–style Industrial AI projects with academic and industry partners, embedding Digital Twin and Predictive Maintenance solutions that improved asset availability by 18–22% for HVAC and rotating equipment.
- Architected end–to–end OT/IT pipelines (sensor → edge → cloud) for high–frequency vibration data, enabling real–time–series analytics, fault classification, and RUL estimation.
- Outlined and deployed edge and MLOps pipelines using Docker and Kubernetes, reducing model deployment and update cycles by 35–40%.
- Developed machine–learning models for high–dimensional industrial data (sensor signals, machine logs), increasing diagnostic precision by 20%+ compared to rule–based approaches.
- Conceptualized Generative AI (LLM, RAG, Agentic AI) decision–support frameworks for maintenance planning and operator guidance in smart manufacturing environments.
- Communicated outcomes to senior stakeholders, engineers, and operators, ensuring clear understanding of operational and business impact.
- Mentored 4 master’s students, delivering 100% of research milestones on schedule using Agile project management practices.

IOT Application Engineer-Consultant

Feb 2024 – Nov 2025

Portainer.io | Remote, Germany

- Served as Technical Program Lead for Industrial IoT and Edge AI consulting engagements across manufacturing and energy clients in the EU.
- Designed scalable OT/IT architectures integrating PLCs, edge gateways, MQTT, and cloud platforms, supporting 10+ production deployments.
- Took ownership of reliable, production–grade software components, designing version–controlled PLC runtimes (CODESYS on WAGO PLCs) and containerized edge services, reducing deployment errors by 30%.
- Advised clients on cloud–based and edge orchestration strategies, improving system availability by 15–20% in high–availability environments.
- Translated complex business and operational requirements into technical solution architectures for Predictive Maintenance, Industrial IoT Analytics, and Digital Twin initiatives.
- Led technical workshops and executive briefings, improving solution adoption and stakeholder alignment by 25%.
- Managed PoC and pilot budgets up to €250k, balancing technical feasibility, scalability, and ROI.

Internship- Autonomous Mobile Robots cloud simulation

Jul 2023 – Jan 2024

Bosch Rexroth | Stuttgart, Germany

- Contributed to cloud-based simulation architectures for Autonomous Mobile Robots (AMRs), improving production planning accuracy by 15%.
- Applied data-driven modelling and simulation to evaluate throughput, layout, and fleet-sizing scenarios in smart manufacturing environments.
- Collaborated with robotics, cloud, and manufacturing teams to validate simulation outputs against real-world constraints.
- Optimized cloud compute utilization, reducing simulation infrastructure cost by ~20% through performance tuning and architectural trade-off analysis.

Working Student- AI Enablement Consultant

Aug 2022 – Jun 2023

Tvarit GmbH, Frankfurt Germany | Frankfurt, Germany

- Supported AI consulting projects focused on process optimization and scrap reduction in manufacturing environments.
- Architected factory data acquisition strategies integrating PLCs, DCS, and energy meters, increasing usable data availability by 30%+.
- Acted as liaison between OEM IT teams for AI Enablement and data scientists, accelerating AI model onboarding by 25%.
- Implemented cloud-based IoT data pipelines (Azure IoT) for scalable ingestion and monitoring of production data.
- Defined and reported technical and business KPIs to senior stakeholders, supporting AI-driven decision-making.

IoT Solution Specialist

Feb 2020 – Dec 2021

Rockwell Automation India (Knowledge Lens) | Bangalore, India

- Led digital twin and predictive maintenance product development for rotary equipment, reducing unplanned downtime by 15% across production sites.
- Outlined and deployed end-to-end Industrial IoT solutions integrating PLCs, edge AI, MES/ERP systems, and cloud analytics platforms for energy forecasting and AI based golden batch reporting.
- Improved fault detection accuracy by 20%+ using AI-based time-series analytics on vibration and sensor data.
- Managed large-scale IoT rollouts across 200+ solar power plants, from shop-floor integration to cloud dashboards.
- Collaborated with plant leadership and engineering teams to improve customer satisfaction and long-term solution adoption by delivering C-level dashboards for production planning and cost optimization.

Core Skills

- Predictive Maintenance, Industrial AI, Digital Twins, Process Optimization, Industrial IoT Analytics, Edge AI, Generative AI (RAG, LLMs)
- Python, Time-Series Analysis, Sensor Data Analytics, Feature Engineering, Model Deployment, High-Dimensional Industrial Data
- OT-IT Architecture, OPC-UA, MQTT, PLCs, SCADA, MES/ERP Integration, Edge Gateways
- Azure IoT, Azure Blob Storage, AWS IoT, Cloud-Based AI Solutions, Data Pipelines, Snowflake

Volunteer Experience

Speaker and Contributor

Oct 2019 – Sep 2020

Open Power Summit 2020

Delivered a talk on Digital Twin based CFD simulator on IBM cloud servers using OpenFoam.

Founding Treasurer

Aug 2019 – Jul 2020

Rotaract Club of NHCE (Rotary)

Lead an accountability of fund raiser for flood relief camp and Blood donation camp at events.

Education

Hochschule Stralsund

Sep 2025 – Present

Master of Arts International Innovation Management
GPA: 2.0

Hochschule Stralsund University of Applied Sciences

Mar 2022 – May 2025

Master of Engineering Simulation and System design (Mechatronics)
Relevant focus: Data Science, Machine Learning, Industrial AI, Systems Engineering
GPA: 1.7

Sep 2019 – Dec 2019

University of Le Havre

Bachelors Exchange Program Energy

Languages

English, German, French (A1), Hindi, Kannada, Marathi

Awards

Aug 2020

Best outgoing Student

New Horizon College of Engineering

Recognized as Best Outgoing Student (R&D) at New Horizon College of Engineering (VTU) for filing two patents and organizing research conferences (ISHRAE, student satellite program).

Certificates

Certified Kubernetes Administrator

KodeKloud and Udemy

Mastering Digital Twin

EIT Digital

Python Programming

Coursera

Open Power Summit

Open Power Foundation

Business Analysis

Udemy

Artificial Intelligence: Agentic AI, Gen AI and RL

Udemy

Publications

NOVEL SYSTEM, METHOD AND DESIGN OF PAPER BAGS MAKING MACHINE WITH INDUSTRY 4.0

Patent office of India Journal

May 2021

NOVEL SYSTEM, METHOD AND DESIGN OF SOLAR PANEL WITH MAXIMUM OUTPUT USING DIGITAL WATCH AND ARDUINO BOARD

Patent office of India Journal

Dec 2021